Future State Journey: Secure Batch Retrieval

Persona: equipment review engineer | Goal: complete a BOM-wide equipment review on schedule without manual lookups

1 2 3 4

1. Submit

USER ACTION

Submit up to 1,000 part numbers in one authenticated request → no sequential loops

INTERNAL STATE

Confident the sanctioned path works → no urge to improvise a workaround

PAIN POINT ADDRESSED

Replaces custom wrappers → no hundreds of sequential GETs

2. Retrieve

USER ACTION

Receive all results in under 1,500 ms → review starts immediately

INTERNAL STATE

Trusts the system is fast → stays focused on engineering judgment

PAIN POINT ADDRESSED

One batch call avoids rate limits → no halfway failures

3. Analyze

USER ACTION

Pass lean, UI-free payload to AI agent → ≥75% fewer tokens

INTERNAL STATE

Empowered, as AI cross-references the BOM → effort shifts to decisions

PAIN POINT ADDRESSED

Strips layout, image URLs, styling → ends payload bloat and token waste

4. Deliver

USER ACTION

Finish BOM-wide equipment review on schedule → deadline met without workarounds

INTERNAL STATE

Secure and compliant → no insecure CSV dumps to explain

PAIN POINT ADDRESSED

Removes scraping and CSV dumps → restores security visibility

Competitive Advantages Over the Manual Workaround

1. Speed

1,000 lookups in under 1,500 ms → replaces hundreds of sequential requests

2. Security

Sanctioned, secure path → closes visibility gaps from scraping and CSV dumps

3. Cost

Stripped, AI-ready payloads → ≥75% lower AI computing cost